

UNIVERSITEIT VAN PRETORIA / UNIVERSITY OF PRETORIA

FAKULTEIT NATUUR- en LANDBOUWETENSKAPPE
FACULTY OF NATURAL and AGRICULTURAL SCIENCES

DEPARTEMENT WISKUNDE EN TOEGEPASTE WISKUNDE
DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Calculus WTW 158
Eksamen / Examination

Interne Eksaminator/ Internal Examiner: Prof JC Engelbrecht

Eksterne Eksaminator/External Examiner: Ms / Me L Mostert

October / Oktober 2006

Tyd / Time: 45 min

Punte / Marks: 30

VAN / SURNAME: _____

VOORNAME / FIRST NAMES: _____

STUDENTENOMMER / STUDENT NUMBER: _____

HANDTEKENING / SIGNATURE: _____

PUNTE
MARKS

LEES DIE VOLGENDE INSTRUKSIES READ THE FOLLOWING INSTRUCTIONS

- | | |
|---|--|
| 1. Die vraestel bestaan uit bladsye 1 tot 4. Kontroleer of jou vraestel volledig is. | 1. The paper consists of pages 1 to 4. Check whether your paper is complete. |
| 2. Doen alle krapwerk op die leë bladsye. Dit word nie nagesien nie. | 2. Do all scribbling on the empty pages. It will not be marked. |
| 3. Toon alle bewerkings. | 3. Show all working details. |
| 4. As jy meer as die beskikbare ruimte vir 'n antwoord nodig het, gebruik dan die leë bladsye en trek 'n raam daarom . | 4. If you need more than the available space for an answer, use the empty pages and please indicate it clearly by framing it. |
| 5. Geen potlood of Tippex nie. | 5. No pencil or Tippex. |

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1. Laat / Let $f(x) = \frac{x-7}{x+7}$.

1.1 Bepaal die horisontale asimptote van f (indien enige). Toon alle berekeninge.

Determine the horizontal asymptotes of f (if any). Show all computations.

[2]

1.2 Bepaal die vertikale asimptote van f (indien enige). Toon alle berekeninge.

Determine the vertical asymptotes of f (if any). Show all computations.

[2]

1.3 Gee die interval(le) waarop f daal. Toon alle berekeninge.

Give the interval(s) where f is decreasing. Show all computations.

[2]

2. Neem aan dat f en g differensieerbare funksies is, waarvan ons die inligting in die tabel beskikbaar het:
Assume that f and g are differentiable functions of which we only know the data in the following table:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	1	4	6

- 2.1 Laat $p(x) = \sqrt{f(x) \cdot g(x) + x^2}$. Bereken $p'(1)$.
 Let $p(x) = \sqrt{f(x) \cdot g(x) + x^2}$ Compute $p'(1)$.

[2]

-
- 2.2 Laat $q(x) = f(g(x))$. Bereken $q'(1)$.
 Let $q(x) = f(g(x))$. Compute $q'(1)$.

[3]

-
3. As / If $f''(x) = \frac{1}{x^3}$, $f(1) = 3$ en / and $f'(1) = -1$, bepaal / find $f(x)$.

4. As $e^y = \ln(x + 2y) + x$, bepaal $\frac{dy}{dx}$ in die punt $(1, 0)$.
 If $e^y = \ln(x + 2y) + x$, determine $\frac{dy}{dx}$ in the point $(1, 0)$.

[2]

[3]

5. Bepaal / Determine

$$\int \frac{x}{\sqrt{3-4x^2}} dx$$

[2]

6. Gegee die volgende parametrisiese vergelykings van twee reguit lyne:
Given the parametric equations of two lines:

$$L_1 : x = 2t, y = 1 - 3t, z = t$$

$$L_2 : x = 1 - 4s, y = 4 + 6s, z = -2s$$

6.1 Toon aan dat L_1 en L_2 ewewydig is.
Show that L_1 and L_2 are parallel.

[2]

6.2 Bepaal 'n Cartesiese vergelyking vir die vlak waarin L_1 en L_2 lê.
Find a Cartesian equation for the plane containing L_1 and L_2 .

[3]

7. Gegee drie punte $A(3, -2, -1)$, $B(0, 4, -2)$ en $C(-5, 2, -3)$.
Given three points $A(3, -2, -1)$, $B(0, 4, -2)$ and $C(-5, 2, -3)$.

7.1 Bepaal die parametrisiese vergelykings van die reguit lyn deur A en B. /
Find the parametric equations of the straight line through A and B.

[2]

7.2 Bepaal die area (oppervlakte) van die driehoek ABC . / *Find the area of the triangle ABC .*

[2]

8. Bepaal die afstand tussen die volgende twee platvlakke:
Determine the distance between the following two planes:

$$\begin{aligned}x - 7y + 8z &= 9 \\- 3x + 21y - 24z &= 3\end{aligned}$$

[3]